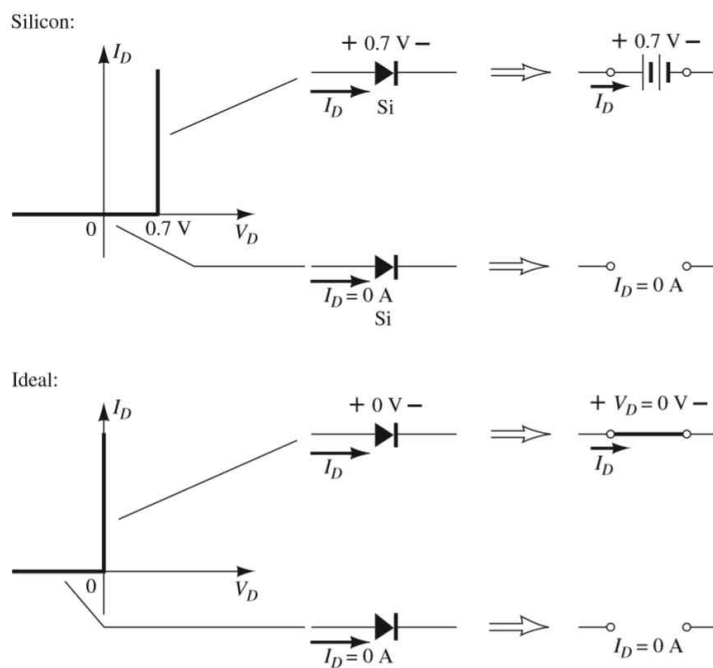


● Series Diode Configurations

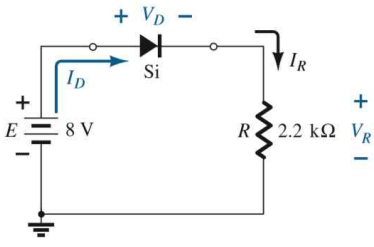
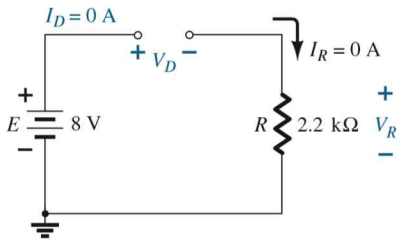
- The forward resistance of the diode is usually so small compared to the other series elements of the network that it can be ignored.
- Approximate and Ideal Semiconductor Diode Models



- Equivalent circuits (Forward / Reverse)

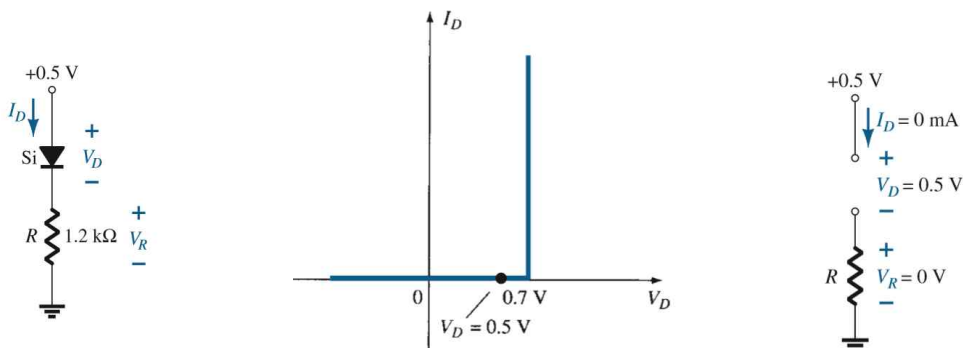
Forward (ON)		(b)
Reverse (OFF)		

(Example 2.4 & 2.5) Determine V_D , V_R and I_D .

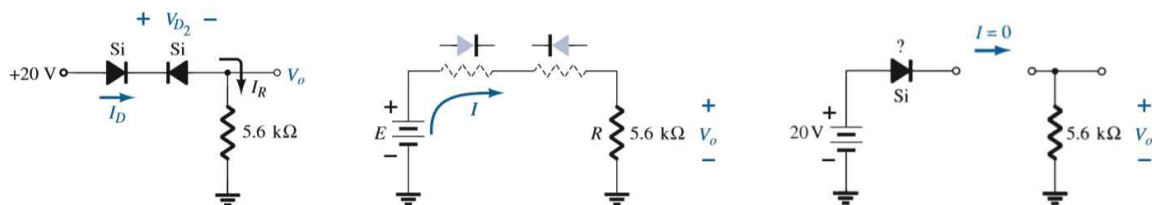
forward bias	reverse bias
	
$V_D = 0.7 \text{ V}$ $V_R = E - V_D = 8 - 0.7 = 7.3 \text{ V}$ $I_D = I_R = \frac{V_R}{R} = \frac{7.3 \text{ V}}{2.2 \text{ k}\Omega} = 3.32 \text{ mA}$	$I_D = 0 \text{ A}$ ($\because$ open circuit) $V_R = I_R R = I_D R = 0 \times 2.2 \text{ k}\Omega = 0 \text{ V}$ KVL $V_D = E - V_R = 8 - 0 = 8 \text{ V}$

※ Example 2.6

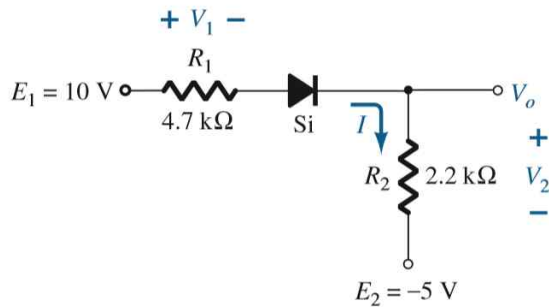
When $E = 0.5 \text{ V}$, what happen?



Example 2.8 Determine I_D , V_{D_2} and V_o .



Example 2.9 Determine I , V_1 , V_2 and V_o .



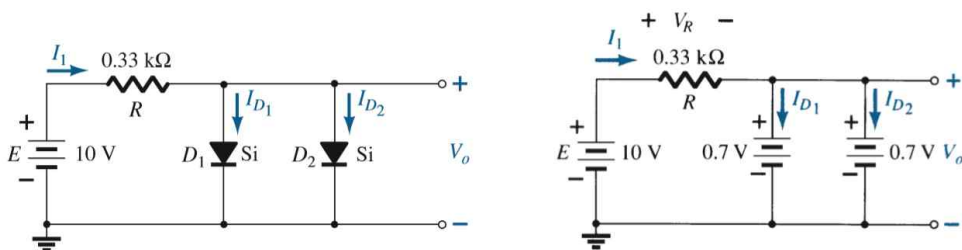
$$I = \frac{E_1 - E_2 - V_D}{R_1 + R_2} = 2.07 \text{ mA}$$

$$V_1 = 9.73 \text{ V} , \quad V_2 = 4.55 \text{ V} ,$$

$$V_2 = V_o - E_2 \quad \therefore V_o = V_2 + E_2 = -0.45 \text{ V}$$

● Parallel and Series-Parallel Diode Configurations

Example 2.10 Determine V_o , I_1 , I_{D_1} and I_{D_2} .



Both diodes are in the 'ON' state.

$$V_o = 0.7 \text{ V}$$

The current is

$$I_1 = \frac{V_R}{R} = \frac{E - V_D}{R} = \frac{10 \text{ V} - 0.7 \text{ V}}{0.33 \text{ k}\Omega} = 28.18 \text{ mA}$$

Assuming diodes of similar characteristics, we have

$$I_{D_1} = I_{D_2} = \frac{I_1}{2} = \frac{28.18 \text{ mA}}{2} = 14.09 \text{ mA}$$

Example 2.13 Determine I_1 , I_2 and I_{D_2} .

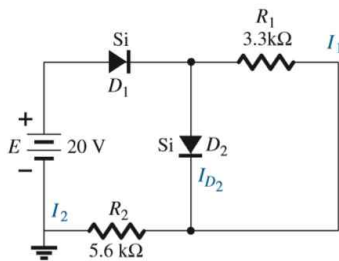


FIG. 37

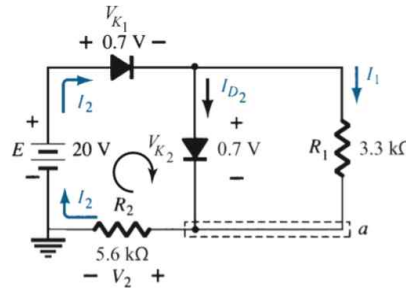


FIG. 38

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series–parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and $V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$

with $I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$

At the bottom node a ,

$$I_{D_2} + I_1 = I_2$$

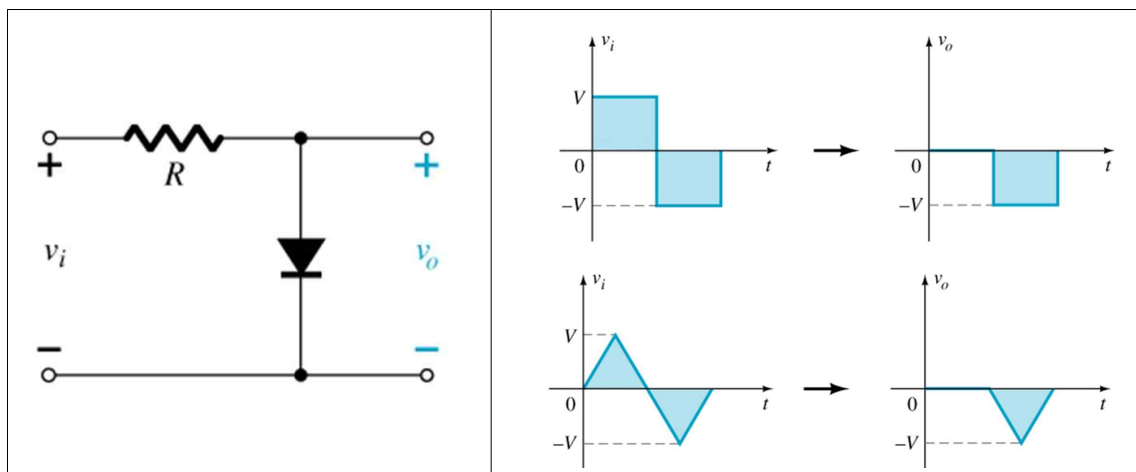
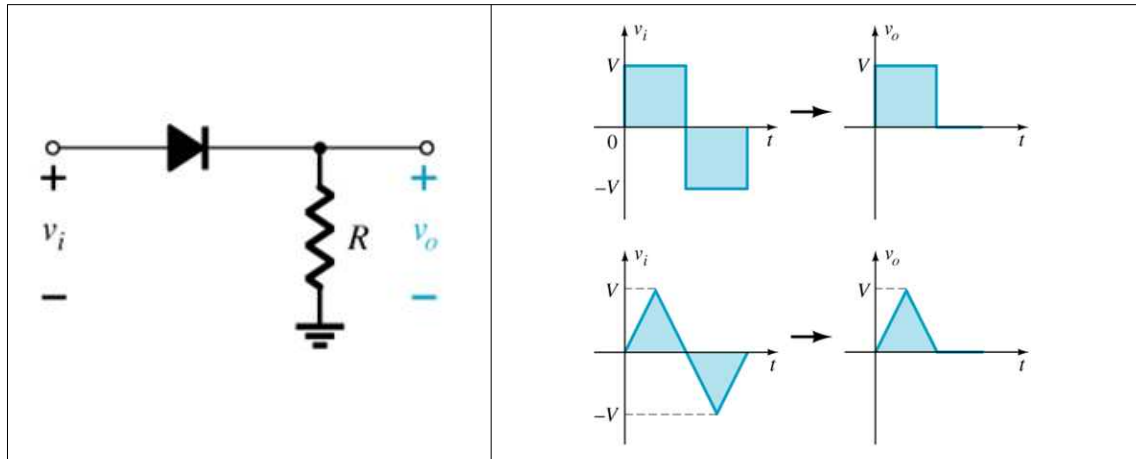
and $I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$

● AND/OR Gates

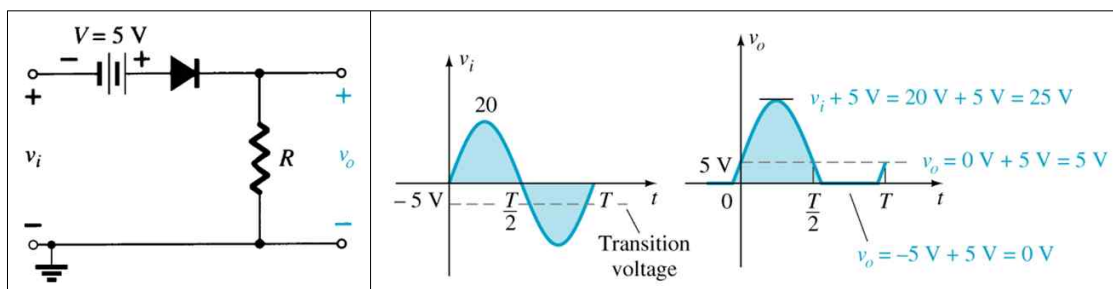
positive logic OR gate	positive logic AND gate

● Clipper Circuits

- Diode Clippers

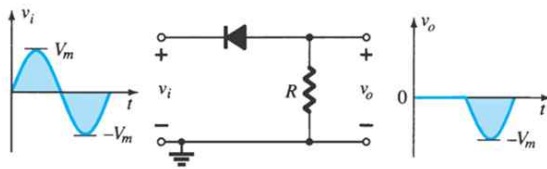


- Biased Clippers

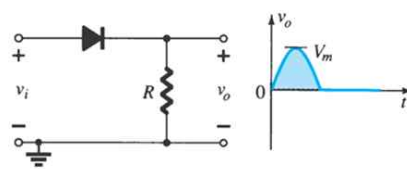


Simple Series Clippers (Ideal Diodes)

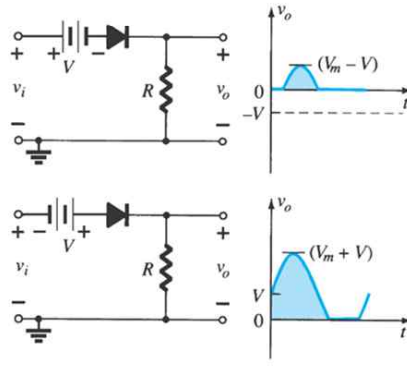
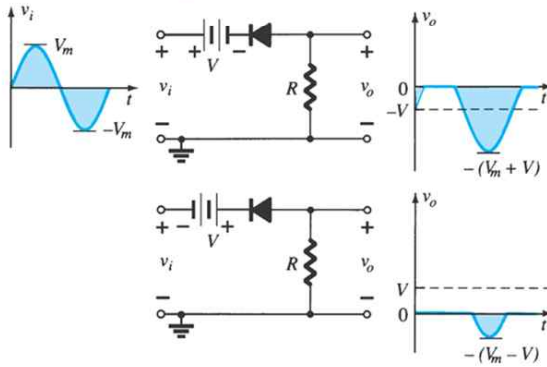
POSITIVE



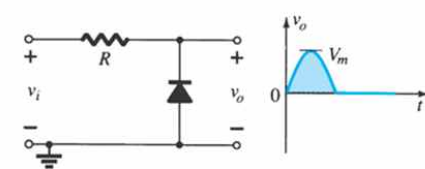
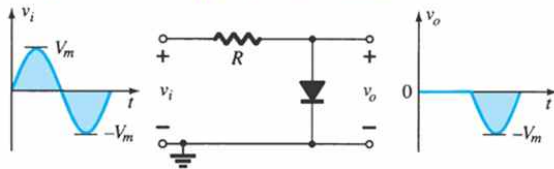
NEGATIVE



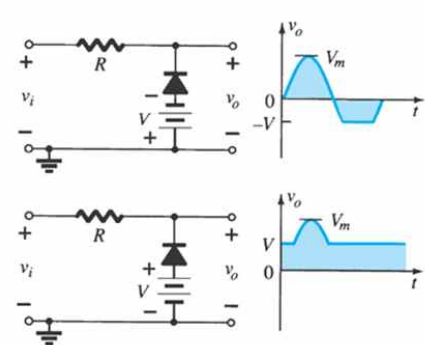
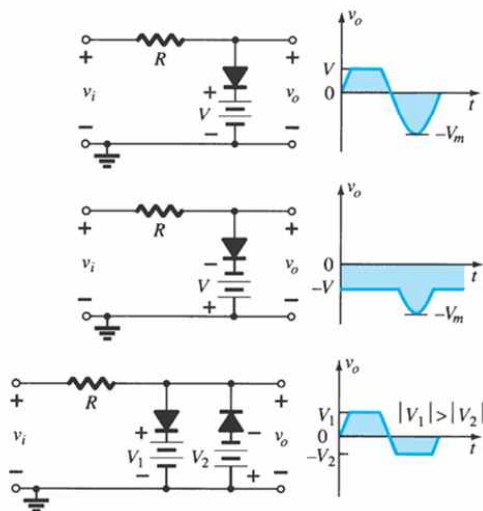
Biased Series Clippers (Ideal Diodes)



Simple Parallel Clippers (Ideal Diodes)

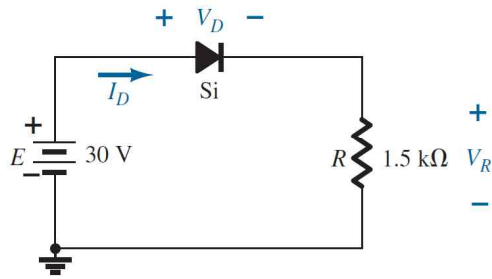


Biased Parallel Clippers (Ideal Diodes)

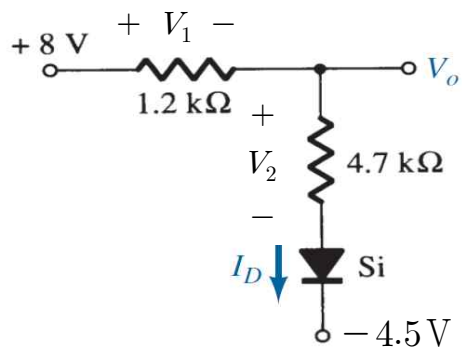


[Examples]

[1] Determine V_D , V_R and I_D .

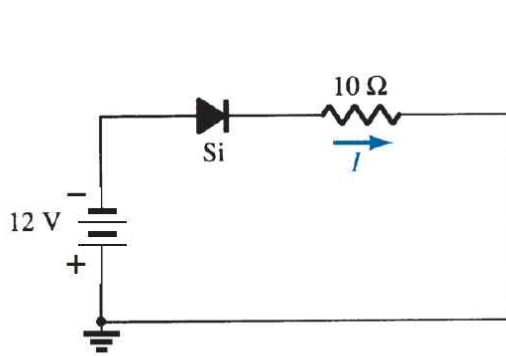


[2] Determine V_1 , V_2 , V_o and I_D .



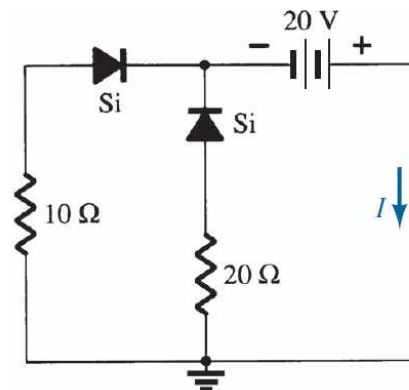
[3] Chap.2 Problem 5 (p.123)

5. Determine the current I for each of the configurations of Fig. 155 using the approximate equivalent model for the diode.



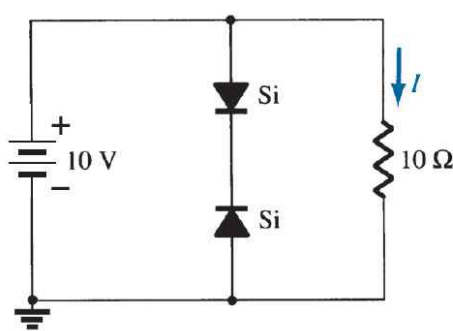
(a)

(a) $I = 0 \text{ mA}$; diode reverse-biased.



(b)

$$\begin{aligned}
 (b) \quad V_{20\Omega} &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(20 \Omega) &= \frac{19.3 \text{ V}}{20 \Omega} = 0.965 \text{ A} \\
 V(10 \Omega) &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(10 \Omega) &= \frac{19.3 \text{ V}}{10 \Omega} = 1.93 \text{ A} \\
 I &= I(10 \Omega) + I(20 \Omega) \\
 &= 2.895 \text{ A}
 \end{aligned}$$



(c)

$$(c) \quad I = \frac{10 \text{ V}}{10 \Omega} = 1 \text{ A}; \text{ center branch open}$$

[4] (Example 2.13)

Determine I_1 , I_2 and I_{D_2} .

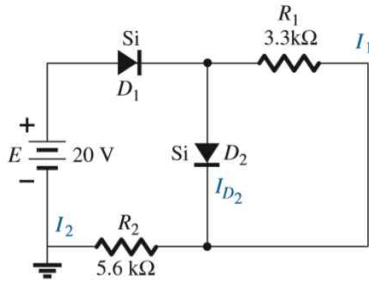


FIG. 37

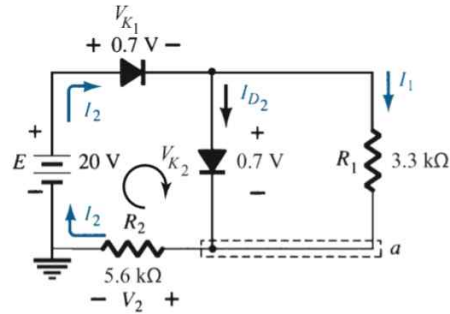


FIG. 38

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series-parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and
$$V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$$

with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$$

At the bottom node a ,

$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$$

[5] Chap.2 Problem 16-17 (p.125)

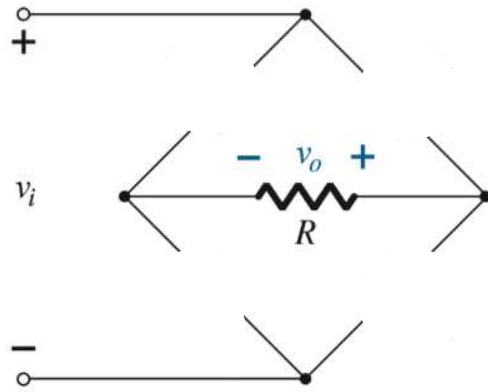
FIG. 42
Positive logic AND gate.

V_1	V_2	V_0
0 V	0 V	
0 V	10 V	
10 V	0 V	
10 V	10 V	
5 V	5 V	

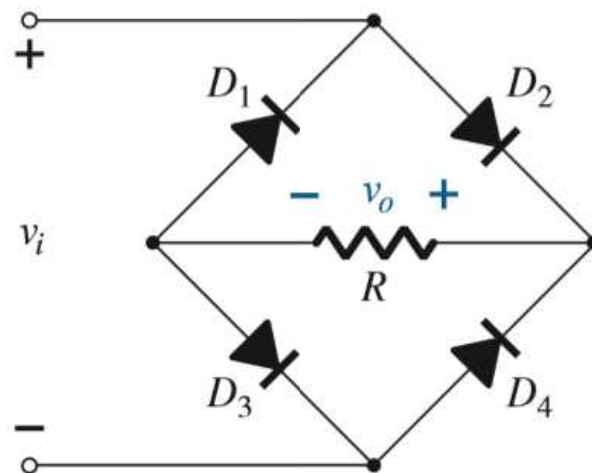
FIG. 39
Positive logic OR gate.

V_1	V_2	V_0
0 V	0 V	
0 V	10 V	
10 V	0 V	
10 V	10 V	
5 V	5 V	

[6] Please make a bridge-type full-wave rectifier using 4 diodes.

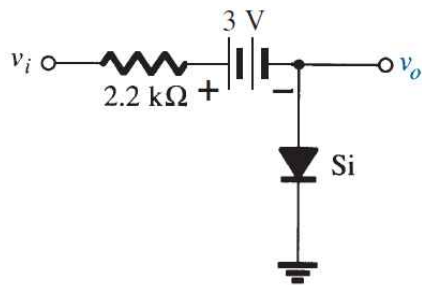
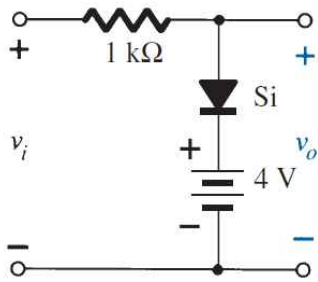


(solution)



[7] Chap.2 Problem 35 (p.128)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .



[8] Chap.2 Problem 34, 36 (p.128)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .

